

A Proof of Shor’s Orthogonal-Measurement Conjecture and the Structure of Information-Optimal Quantum Measurements

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Which quantum measurement extracts the most classical information from an ensemble? We introduce the posterior algebra, a new canonical operator algebra selected by mutual information. For faithful ensembles, an affine information bound is exact precisely when this algebra is commutative; its joint spectral measurement is then optimal, and every optimal finite POVM refines it. Binary ensembles have one generator; compactness covers singular states, giving a proof of Shor’s finite-dimensional binary orthogonal-measurement conjecture. The framework also gives rigidity bounds and a certified posterior-spectral receiver, validated on 408 mixed-state instances.

Quantum measurement design depends on the operational objective. The Helstrom measurement minimizes the probability of a wrong hard decision [1]. A receiver for communication, soft decoding, or statistical inference should instead maximize the mutual information between the signal label and the measurement outcome. These objectives are different. For more than two signals, generalized measurements can strictly outperform every projective measurement [2–4]. Even for two mixed states, the minimum-error and maximum-information bases can differ.

Early binary results include ensemble-dependent information bounds and Levitin’s proof for two pure states [5, 6]. Shor later refuted Levitin’s broader conjecture while reporting Fuchs and Peres’s numerical studies, in which the optimal binary measurement was always orthogonal [3]. This surviving statement became known as Shor’s orthogonal-measurement conjecture; Keil proved the qubit case [7]. As late as December 2025, Thai and Dall’Arno still described the general statement as an open conjecture; they disproved a proposed monotonicity route and characterized state-dependent extremality for qubit dichotomies [8]. However, we also observe that the projective-reduction theorem of Fang, Fawzi, and Fawzi, first posted in February 2025, already implies the binary—and, more generally, collinear—existence statement after the specialization verified in Appendix D [9]. We therefore claim no priority for projective sufficiency itself. Our route instead exposes the optimizer-selected algebra, classifies every optimal finite POVM in the exact regime, and supplies quantitative and algorithmic consequences.

Let $\mathcal{E} = \{(p_i, \rho_i)\}_{i=1}^m$ be a finite ensemble with $p_i > 0$ (zero-prior signals having been removed), and let $\tau = \sum_i p_i \rho_i$ be its average state. A POVM $M = \{M_y\}_y$ produces $q(y|i) = \text{Tr } \rho_i M_y$ and $q(y) = \text{Tr } \tau M_y$. The accessible information is

$$I(\mathcal{E}, M) = \sum_{i,y} p_i q(y|i) \log \frac{q(y|i)}{q(y)},$$

$$I_{\text{acc}}(\mathcal{E}) = \sup_M I(\mathcal{E}, M). \quad (1)$$

The optimization is nonlinear and ranges over all generalized measurements. Davies’ theorem bounds the required number of effects by d^2 , but it does not identify the optimizer [10]. The basic questions therefore remain structural. When does a projective measurement suffice? Is there a canonical optimum? How are all optimal measurements related? Can an answer also lead to a certified algorithm?

Our results.— For faithful finite ensembles, we answer these questions by associating an operator algebra with the information task itself. We first optimize an ensemble-adapted affine upper bound U_{aff} . Its unique optimizer defines posterior operators G_i^* and the *posterior algebra*

$$\mathcal{A}_\star = C^*(G_1^*, \dots, G_m^*). \quad (2)$$

Our central result is

$$I_{\text{acc}} = U_{\text{aff}} \iff \mathcal{A}_\star \text{ is abelian.} \quad (3)$$

In the exact regime, the joint spectral PVM of $\mathcal{A}_\star$ is canonical. Every optimal finite POVM is a label-independent refinement of this PVM. We also prove a quantitative form of the statement. If $\epsilon_M = U_{\text{aff}} - I(\mathcal{E}, M)$, then an optimized block-intertwining residual obeys

$$R_\star(M) \leq 4\epsilon_M. \quad (4)$$

In the abelian exact regime, near-optimal measurements must therefore nearly respect the same posterior blocks. For a binary ensemble the affine dimension is one, so the posterior algebra has one generator and is automatically abelian.

Equation (3) then proves Shor’s orthogonal-measurement conjecture. The same reduction yields a certified posterior-spectral receiver. It constructs a projective basis and bounds how much additional mutual information any other measurement could gain. Figure 1 compares its finite-budget accuracy and runtime profiles with standard receiver-design methods.

Posterior algebra.— Assume first that every signal state

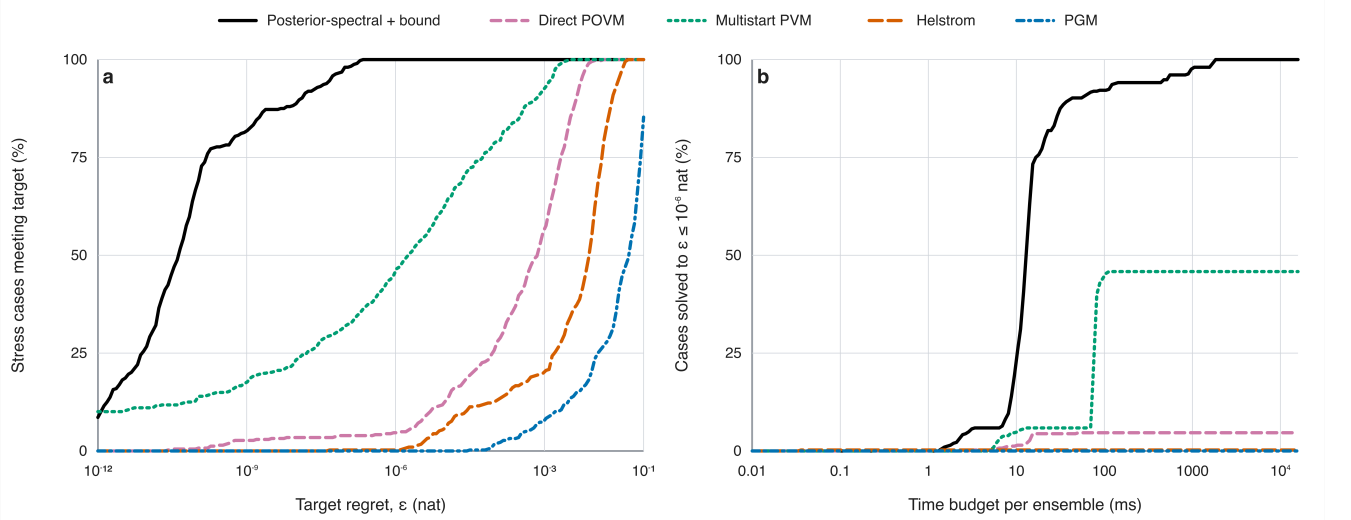


FIG. 1. Performance of the posterior-spectral receiver on 408 faithful binary mixed-state stress instances. (a) Fraction of instances whose information regret is below a target ϵ . (b) Fraction solved to $\epsilon \leq 10^{-6}$ nat within a given time budget. Higher and further left are better. The posterior-spectral curve uses its certified affine upper-minus-PVM gap. The other curves use empirical regret relative to the posterior-spectral PVM.

is faithful. Choose minimal real affine coordinates

$$\rho_i = \tau + \sum_{a=1}^r s_{ia} A_a, \quad \sum_i p_i s_{ia} = 0. \quad (5)$$

For a Hermitian tuple $\mathbf{H} = (H_1, \dots, H_r)$, define

$$\begin{aligned} G_i(\mathbf{H}) &= I + \sum_a s_{ia} H_a \succ 0, \\ \Phi(\mathbf{H}) &= \sum_i p_i \text{Tr} \rho_i \log G_i(\mathbf{H}), \\ U_{\text{aff}} &= \max_{\mathbf{H}} \Phi(\mathbf{H}). \end{aligned} \quad (6)$$

The feasible domain is bounded, and Φ tends to $-\infty$ at its boundary. Intrinsic strong concavity gives a unique interior maximizer $\mathbf{H}_*$. The resulting operators $G_i^* = G_i(\mathbf{H}_*)$ are quantum analogues of posterior likelihood ratios. A change of affine coordinates leaves their generated algebra unchanged.

The program bounds every physical measurement. For each nonzero POVM effect (so $c_y > 0$ by faithfulness of τ), set

$$\begin{aligned} c_y &= \text{Tr} \tau M_y, & h_{ay} &= \frac{\text{Tr} A_a M_y}{c_y}, \\ H_{a,M} &= \sum_y h_{ay} M_y, & g_{iy} &= \frac{q(y|i)}{c_y}. \end{aligned} \quad (7)$$

Then $G_i(\mathbf{H}_M) = \sum_y g_{iy} M_y$. Operator Jensen gives

$$\begin{aligned} \log G_i(\mathbf{H}_M) &\succeq \sum_y (\log g_{iy}) M_y, \\ I(\mathcal{E}, M) &\leq \Phi(\mathbf{H}_M) \leq U_{\text{aff}}. \end{aligned} \quad (8)$$

Hence $I_{\text{acc}} \leq U_{\text{aff}} \leq \chi$, where χ is the Holevo quantity [11–15]. The gap $U_{\text{aff}} - I_{\text{acc}}$ measures the cost of incompatible optimal posterior observables.

We now prove the exactness criterion. Suppose first that the optimized posteriors commute. Let $P_* = \{P_z\}_z$ be their minimal joint-spectral PVM, with $G_i^* P_z = g_{iz} P_z$. Direct expansion gives

$$I(\mathcal{E}, P_*) - \Phi(\mathbf{H}_*) = \sum_z \text{Tr}(\tau P_z) D(\alpha_{\cdot z} \| \beta_{\cdot z}) \geq 0, \quad (9)$$

where $\alpha_{iz} = p_i \text{Tr}(\rho_i P_z) / \text{Tr}(\tau P_z)$ and $\beta_{iz} = p_i g_{iz}$. Since $I(\mathcal{E}, P_*) \leq I_{\text{acc}} \leq U_{\text{aff}} = \Phi(\mathbf{H}_*)$, every inequality is an equality. Thus P_* is optimal.

Conversely, suppose $I_{\text{acc}} = U_{\text{aff}}$. An optimal finite POVM must saturate both inequalities in Eq. (8). Uniqueness gives $\mathbf{H}_M = \mathbf{H}_*$. Equality in compression Jensen then implies

$$g_{iy} M_y^{1/2} = M_y^{1/2} G_i^*. \quad (10)$$

Every nonzero effect is supported in a common eigenspace of all G_i^* . Completeness therefore forces $\mathcal{A}_*$ to be abelian. The stationarity equations also give $P_z \rho_i P_z = g_{iz} P_z \tau P_z$. It follows that a finite POVM is optimal if and only if its nonzero outcomes can be partitioned as

$$M_y = P_z M_y P_z \quad (y \in Y_z), \quad \sum_{y \in Y_z} M_y = P_z. \quad (11)$$

The refinement channel is independent of the signal label. All optimal finite POVMs are therefore Blackwell-equivalent to P_* [16, 17]; see Appendix B.

Proof of Shor's conjecture. —

Corollary 1 (Shor’s conjecture). *For every finite-dimensional binary ensemble, accessible information is attained by a rank-one projective measurement.*

Proof. Let $\mathcal{E} = \{(p, \rho_0), (q, \rho_1)\}$, with $q = 1 - p$. The cases $pq = 0$ or $\rho_0 = \rho_1$ are immediate, so assume otherwise. Put $A = \rho_0 - \rho_1$. Since $\rho_0 = \tau + qA$ and $\rho_1 = \tau - pA$, the affine program has one Hermitian variable and

$$G_0^* = I + qH_*, \quad G_1^* = I - pH_*. \quad (12)$$

Both are functions of H_* , so they commute. For faithful states, Eq. (3) therefore makes the spectral PVM of H_* optimal. Refining a degenerate spectral block into rank-one projections cannot decrease mutual information; since the original PVM is optimal, the refinement is optimal as well.

For singular states let $P_0 = \text{supp } \tau$. Positivity of p, q implies $\text{supp } \rho_i \subseteq P_0$. On $P_0\mathcal{H}$, set $\rho_{i,\epsilon} = (1 - \epsilon)\rho_i + \epsilon P_0/d_0$, where $d_0 = \text{rank } P_0$. The faithful case supplies an ordered optimal rank-one PVM $P^{(\epsilon)}$. Compactness of ordered rank-one PVMs gives a subsequence $P^{(\epsilon_k)} \rightarrow P$ as $\epsilon_k \downarrow 0$. Joint continuity of finite-alphabet mutual information implies, for every fixed finite POVM M ,

$$I(\mathcal{E}, P) = \lim_k I(\mathcal{E}_{\epsilon_k}, P^{(\epsilon_k)}) \geq \lim_k I(\mathcal{E}_{\epsilon_k}, M) = I(\mathcal{E}, M). \quad (13)$$

Davies’ finite-outcome attainment theorem makes P globally optimal. A rank-one resolution of $P_0^\perp$ extends it to the original Hilbert space without changing any signal statistics; see Appendix D. $\square$

This proof also identifies the canonical measurement, classifies every optimizer in the faithful exact regime, and yields rigidity and a constructive algorithm. The prior measured- f route and its exact scope are verified in Appendix D.

Posterior-spectral receiver.— For a faithful binary ensemble, the receiver takes p, ρ_0 , and ρ_1 as input. It returns a rank-one projective measurement P and an interval $[L, U]$ such that $L = I(\mathcal{E}, P) \leq I_{\text{acc}} \leq U$. Thus $U - L$ answers a practical question: how much more information could any other measurement still extract? The method avoids a direct search over many POVM effects or projective bases. Instead, it solves one strictly concave problem for a Hermitian operator H :

$$H_* = \arg \max_H \{p \text{Tr } \rho_0 \log(I + qH) + q \text{Tr } \rho_1 \log(I - pH)\}, \quad (14)$$

subject to $I + qH \succ 0$ and $I - pH \succ 0$. At the optimum, the receiver is the spectral PVM of H_* . At any feasible iterate $\hat{H}$, the spectral PVM $P_{\hat{H}}$ gives the lower bound, while the intrinsic gradient residual r_g gives

$$I(\mathcal{E}, P_{\hat{H}}) \leq I_{\text{acc}} \leq U_{\text{aff}} \leq \Phi(\hat{H}) + \frac{1}{2}r_g^2. \quad (15)$$

The upper-minus-lower gap is therefore a verifiable stopping criterion. Standard direct-POVM and nonconvex-PVM optimizers provide only lower bounds.

Algorithm 1 — Posterior-spectral receiver

Input: $\mathcal{E} = \{(p, \rho_0), (q, \rho_1)\}$, $q = 1 - p$, with faithful states; target gap $\varepsilon > 0$. *Output:* a rank-one PVM P_H and a certificate $0 \leq I_{\text{acc}} - I(\mathcal{E}, P_H) \leq \varepsilon$.

1. Set $H \leftarrow 0$ and $C = p^2 q^2 [p \lambda_{\min}(\rho_0) + q \lambda_{\min}(\rho_1)]$.
 2. Form $G_0 = I + qH$ and $G_1 = I - pH$. Evaluate $\Phi(H)$ and

$$R(H) = pq [\mathcal{L}_{G_0}(\rho_0) - \mathcal{L}_{G_1}(\rho_1)], \quad r_g^2 = \text{Tr } R(H)^2 / C,$$

where $\mathcal{L}_G = D \log G$.

3. Diagonalize $H = \sum_z h_z P_z$, refine degeneracies to rank one, and evaluate $L = I(\mathcal{E}, P_H)$ and $U = \Phi(H) + r_g^2/2$.

4. If $U - L \leq \varepsilon$, return P_H and the certificate $0 \leq I_{\text{acc}} - L \leq U - L$.

5. Otherwise take a feasibility-preserving ascent step for Φ and return to Step 2.

The stress suite in Fig. 1 spans dimensions 2, 4, 8, 16, spectral floors down to 10^{-6} , priors up to 0.99, and nearly coincident state pairs. With the stated fixed iteration budgets and float64 evaluations of the analytic bound, the posterior-spectral receiver reaches 10^{-6} nat on all 408 instances. Multistart PVM and direct POVM ascent reach the same target on 45.8% and 4.7% of the instances. The complete protocol is in Appendix E.

Near-optimal measurements.— The certificate controls the achieved information value. We next ask whether a small gap also constrains the measurement itself. For faithful binary ensembles, $U_{\text{aff}} = I_{\text{acc}}$, so $\epsilon_M = U_{\text{aff}} - I(\mathcal{E}, M)$ is the true information regret of any POVM M . Let $a_i = p_i^3 \lambda_{\min}(\rho_i)$ and define

$$R_*(M) = \sum_{i,y} a_i \|g_{iy} M_y^{1/2} - M_y^{1/2} G_i^*\|_F^2. \quad (16)$$

Each term vanishes when outcome y lies in a posterior spectral block and has the corresponding posterior ratio. A quantitative Jensen remainder and strong concavity give

$R_*(M) \leq 4\epsilon_M.$

(17)

The information gap therefore controls a concrete structural error. More explicitly, write $G_i^* = \sum_z g_{iz} P_z$ and let Δ_{post} be the minimum weighted separation between distinct posterior eigenvalue vectors. Assign each outcome to its dominant block and let $L(M)$ be the total weight outside the assigned blocks. Then

$$L(M) \leq \frac{8\epsilon_M}{\Delta_{\text{post}}^2}. \quad (18)$$

Thus every near-optimal measurement is close, as a statistical experiment, to a label-independent refinement of the canonical posterior PVM. This is the bridge from an objective-value certificate to a device-level diagnostic; see Appendix C.

Receiver-design simplification.— For binary finite-dimensional ensembles, our result gives an exact reduction of the single-copy receiver-design problem. A generic optimization may involve up to d^2 nonorthogonal POVM effects and, in hardware, an ancilla-assisted realization. Corollary 1 reduces this search without loss of accessible information to an orthogonal measurement on the d -dimensional signal space. For faithful states, the posterior-spectral construction sharpens the reduction further: instead of optimizing the effects or performing a nonconvex search over analyzer bases, one solves the concave program in Eq. (14) and diagonalizes a single Hermitian operator. The resulting receiver requires only a programmable basis rotation followed by fixed orthogonal readout, while Eq. (15) certifies the maximum information gain left to any alternative single-copy measurement. Moreover, in this faithful regime every optimal finite POVM is a label-independent refinement of the posterior-spectral PVM, so outcomes within the same posterior block may be merged without information loss. The result can therefore simplify receiver optimization, ancilla and detector requirements, readout dimensionality, and calibration, provided that the task is binary, single-copy, and optimized for mutual information.

Conclusion and outlook.— Information-optimal readout is governed by a task-specific posterior algebra. Its commutativity decides whether the affine bound is attainable, and its spectral blocks organize every optimal finite measurement. With one generator, binary ensembles turn Shor’s conjecture into a structural consequence and reduce receiver design to a certified concave program plus spectral readout. Varying-support rigidity, validated numerics, and collective measurements remain open.

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Data and code availability.— No experimental data were generated. Machine-readable benchmark records, deterministic checks, analysis scripts, and figure-generation code are included in the `anc/` directory of the arXiv source archive.

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Appendix A: Affine geometry and the information upper bound

Throughout this section, $p_i > 0$ and every ρ_i is faithful. Let $V = \text{span}_{\mathbb{R}}\{\rho_i - \tau\}$ and choose a minimal representation

$$\rho_i = \tau + \sum_{a=1}^r s_{ia} A_a, \quad \sum_i p_i s_{ia} = 0, \quad G_i(\mathbf{H}) = I + \sum_a s_{ia} H_a \succ 0. \quad (\text{A1})$$

The feasible domain is nonempty and convex. Since $\sum_i p_i G_i(\mathbf{H}) = I$, every feasible posterior satisfies $G_i \preceq p_i^{-1} I$. Full column rank of the coordinate matrix bounds $\mathbf{H}$. If a feasible sequence approaches the boundary, some G_j acquires an eigenvalue tending to zero. Faithfulness of ρ_j implies $\text{Tr } \rho_j \log G_j \rightarrow -\infty$, while all other terms remain bounded above. The maximum is therefore attained in the interior.

For $X \succ 0$, $R \succ 0$, and Hermitian K , operator concavity of the logarithm and its integral representation give

$$\frac{\lambda_{\min}(R)}{\|X\|_{\infty}^2} \|K\|_F^2 \leq -D^2 \text{Tr } R \log X[K, K] \leq \frac{\|R\|_{\infty}}{\lambda_{\min}(X)^2} \|K\|_F^2. \quad (\text{A2})$$

For a tuple direction $\mathbf{K}$, put $K_i = \sum_a s_{ia} K_a$. Since $G_i \preceq p_i^{-1} I$, Eq. (A2) yields

$$-D^2 \Phi(\mathbf{H})[\mathbf{K}, \mathbf{K}] \geq \sum_i p_i^3 \lambda_{\min}(\rho_i) \|K_i\|_F^2. \quad (\text{A3})$$

Minimality of the coordinates makes the right side a norm, so Φ is strongly concave and its interior optimizer is unique. Its stationarity conditions are

$$\begin{aligned} \sum_i p_i s_{ia} \mathcal{L}_{G_i^*}(\rho_i) &= 0, \\ \mathcal{L}_F(X) &= D \log F[X] = \int_0^\infty (F + uI)^{-1} X (F + uI)^{-1} du. \end{aligned} \quad (\text{A4})$$

For a finite POVM without zero effects, set

$$\begin{aligned} c_y &= \text{Tr } \tau M_y, & h_{ay} &= \frac{\text{Tr } A_a M_y}{c_y}, \\ H_{a,M} &= \sum_y h_{ay} M_y, & g_{iy} &= \frac{\text{Tr } \rho_i M_y}{c_y}. \end{aligned} \quad (\text{A5})$$

Then

$$G_i(\mathbf{H}_M) = \sum_y g_{iy} M_y. \quad (\text{A6})$$

The map $X \mapsto \log X$ is operator concave, hence

$$\log G_i(\mathbf{H}_M) \succeq \sum_y (\log g_{iy}) M_y. \quad (\text{A7})$$

Tracing against $p_i \rho_i$, summing over i , and using the definition of g_{iy} proves $I(\mathcal{E}, M) \leq \Phi(\mathbf{H}_M) \leq U_{\text{aff}}$.

For completeness, introduce the unrestricted value

$$U_{\text{unres}} = \max_{F_i \succ 0, \sum_i p_i F_i = I} \sum_i p_i \text{Tr } \rho_i \log F_i. \quad (\text{A8})$$

Affine feasibility gives $U_{\text{aff}} \leq U_{\text{unres}}$. For an unrestricted feasible family, set $q_i = \text{Tr } \tau F_i$ and $Q_i = F_i/q_i$. Then $\text{Tr } \tau Q_i = 1$, and the variational formula for measured relative entropy implies

$$\text{Tr } \rho_i \log F_i \leq D_{\mathbb{M}}(\rho_i \| \tau) + \log q_i. \quad (\text{A9})$$

Moreover, $\sum_i p_i q_i = 1$, so scalar Jensen gives $\sum_i p_i \log q_i \leq 0$. Consequently,

$$I_{\text{acc}} \leq U_{\text{aff}} \leq U_{\text{unres}} \leq \sum_i p_i D_{\mathbb{M}}(\rho_i \| \tau) \leq \chi(\mathcal{E}). \quad (\text{A10})$$

The variational and equality statements used here are standard [12, 13, 15]. This endpoint is not needed for the posterior-algebra exactness theorem, but it places U_{aff} between the difficult nonlinear POVM optimization and the explicit Holevo quantity.

Appendix B: Posterior exactness and all optimal measurements

We first record equality in compression Jensen. Let $V : \mathcal{H} \rightarrow \mathcal{K}$ be an isometry, $T \succ 0$, and $G = V^*TV$. In the decomposition $\text{ran } V \oplus (\text{ran } V)^\perp$, write

$$T = \begin{pmatrix} G & B \\ B^* & C \end{pmatrix}. \quad (\text{B1})$$

The integral representation of the logarithm and the Schur complement yield

$$\log G - V^*(\log T)V = \int_0^\infty [(G + uI - B(C + uI)^{-1}B^*)^{-1} - (G + uI)^{-1}] du \succeq 0. \quad (\text{B2})$$

Equality holds if and only if $B = 0$, equivalently $TV = VG$. For the POVM dilation

$$Vx = \bigoplus_y M_y^{1/2}x, \quad T_i = \bigoplus_y g_{iy}I, \quad (\text{B3})$$

this gives

$$g_{iy}M_y^{1/2} = M_y^{1/2}G_i \quad \text{for every } i, y. \quad (\text{B4})$$

Suppose first that the optimized posteriors commute. Let $P_\star = \{P_z\}_z$ be their minimal joint-spectral PVM and write $G_i^*P_z = g_{iz}P_z$. Define

$$\alpha_{iz} = \frac{p_i \text{Tr}(\rho_i P_z)}{\text{Tr}(\tau P_z)}, \quad \beta_{iz} = p_i g_{iz}. \quad (\text{B5})$$

Both are probability vectors for each z , and a direct expansion gives

$$I(\mathcal{E}, P_\star) - \Phi(\mathbf{H}_\star) = \sum_z \text{Tr}(\tau P_z) D(\alpha_{\cdot z} \| \beta_{\cdot z}) \geq 0. \quad (\text{B6})$$

Since $I(\mathcal{E}, P_\star) \leq I_{\text{acc}} \leq U_{\text{aff}} = \Phi(\mathbf{H}_\star)$, equality holds throughout and $P_\star$ is optimal.

Conversely, assume $I_{\text{acc}} = U_{\text{aff}}$. Finite-dimensional attainment supplies an optimal finite POVM M . Equality in

$$I(\mathcal{E}, M) \leq \Phi(\mathbf{H}_M) \leq \Phi(\mathbf{H}_\star) \quad (\text{B7})$$

forces $\mathbf{H}_M = \mathbf{H}_\star$ by uniqueness. The positive Jensen defect has zero trace against each faithful state and is therefore zero as an operator. Equation (B4) holds with $G_i = G_i^*$. The support of every nonzero effect lies in a common eigenspace of all G_i^* ; POVM completeness makes these supports span $\mathcal{H}$. Hence $\mathcal{A}_\star$ is abelian.

It remains to classify the optimizers. Compress Eq. (A4) to a joint spectral block P_z and put

$$T_z = P_z \tau P_z, \quad B_{az} = P_z A_a P_z, \quad C_z = \sum_i \frac{p_i}{g_{iz}} s_i s_i^\top. \quad (\text{B8})$$

The matrix C_z is positive definite. If $h_z = (h_{1z}, \dots, h_{rz})^\top$, normalization gives $\sum_i (p_i/g_{iz}) s_i + C_z h_z = 0$, while the compressed stationarity equations give $C_z(B_z - h_z T_z) = 0$. Therefore

$$B_{az} = h_{az} T_z, \quad P_z \rho_i P_z = g_{iz} P_z \tau P_z. \quad (\text{B9})$$

Equation (B4) shows that every optimal effect is supported in one block. Conversely, any block refinement satisfies

$$\text{Tr} \rho_i M_y = g_{iz} \text{Tr} \tau M_y, \quad y \in Y_z. \quad (\text{B10})$$

Thus the conditional refinement channel

$$R(y|z) = \frac{\text{Tr}(\tau M_y)}{\text{Tr}(\tau P_z)} \quad (\text{B11})$$

is independent of the label i , while $y \mapsto z$ is deterministic. The induced classical experiments are Blackwell-equivalent [16, 17].

Appendix C: Robust posterior rigidity

We now quantify the equality mechanism. If $0 \prec T \preceq \beta I$, the block representation used in Eq. (B2) and the resolvent identity imply

$$\text{Tr}[\log G - V^*(\log T)V] \geq \frac{1}{2\beta^2} \|TV - VG\|_F^2. \quad (\text{C1})$$

Indeed, writing $X_u = G + uI$ and $S_u = B(C + uI)^{-1}B^*$, the trace of the integrand is bounded below by $(\beta + u)^{-3} \|B\|_F^2$. Integration gives the stated constant and $\|B\|_F = \|TV - VG\|_F$. If $\rho \succeq \mu I$, positivity of the defect gives the corresponding lower bound after tracing against ρ .

Define the intrinsic posterior norm

$$\|K\|_{\text{post}}^2 = \sum_i a_i \|K_i\|_F^2, \quad a_i = p_i^3 \lambda_{\min}(\rho_i). \quad (\text{C2})$$

Equation (A3) says that Φ is one-strongly concave in this norm. Split the information gap as

$$\epsilon_M = U_{\text{aff}} - I(\mathcal{E}, M) = \epsilon_{\text{opt}} + \epsilon_J, \quad \begin{cases} \epsilon_{\text{opt}} = \Phi(\mathbf{H}_\star) - \Phi(\mathbf{H}_M), \\ \epsilon_J = \Phi(\mathbf{H}_M) - I(\mathcal{E}, M). \end{cases} \quad (\text{C3})$$

Strong concavity gives $\sum_i a_i \|G_i(\mathbf{H}_M) - G_i^*\|_F^2 \leq 2\epsilon_{\text{opt}}$. For the POVM dilation, $T_i \preceq p_i^{-1}I$ because $g_{iy} = q(y|i)/q(y) \leq p_i^{-1}$. Applying Eq. (C1) and then tracing against $p_i \rho_i$ gives the corresponding residual relative to $G_i(\mathbf{H}_M)$ bounded by $2\epsilon_J$. Minkowski's inequality in the weighted direct sum then yields

$$R_\star(M) = \sum_{i,y} a_i \|g_{iy} M_y^{1/2} - M_y^{1/2} G_i^*\|_F^2 \leq 4\epsilon_M. \quad (\text{C4})$$

No commutativity assumption enters this estimate.

When the optimized posterior algebra is abelian, write $G_i^* = \sum_z g_{iz} P_z$ and define

$$\Delta_{\text{post}}^2 = \min_{z \neq z'} \sum_i a_i (g_{iz} - g_{iz'})^2. \quad (\text{C5})$$

Write $g_z = (g_{1z}, \dots, g_{mz})$, $g_y = (g_{1y}, \dots, g_{my})$, and $\|v\|_a^2 = \sum_i a_i v_i^2$. For an outcome y , expansion over the spectral blocks gives

$$\sum_i a_i \|M_y^{1/2} G_i^* - g_{iy} M_y^{1/2}\|_F^2 = \sum_z \|g_z - g_y\|_a^2 \text{Tr } P_z M_y. \quad (\text{C6})$$

Lemma 2 (Weighted separation). *Let $u_1, \dots, u_K$ obey $\|u_z - u_{z'}\|_a^2 \geq \Delta^2$ for $z \neq z'$, and let $w_z \geq 0$, $W = \sum_z w_z$. For every u ,*

$$\sum_z w_z \|u_z - u\|_a^2 \geq \frac{\Delta^2}{2} (W - \max_z w_z). \quad (\text{C7})$$

Proof. The left side is minimized at $\bar{u} = W^{-1} \sum_z w_z u_z$. The weighted variance identity gives

$$\sum_z w_z \|u_z - \bar{u}\|_a^2 = \frac{1}{2W} \sum_{z, z'} w_z w_{z'} \|u_z - u_{z'}\|_a^2. \quad (\text{C8})$$

The off-diagonal terms are at least Δ^2 , and $\sum_z w_z^2 \leq W \max_z w_z$, proving Eq. (C7). $\square$

Assign y to a block $z(y)$ maximizing $\text{Tr } P_z M_y$. Apply Lemma 2 with $w_z = \text{Tr } P_z M_y$, sum over outcomes, and use Eq. (C4). This proves the wrong-block mass bound

$$L(M) = \sum_y \sum_{z \neq z(y)} \text{Tr } P_z M_y \leq \frac{8\epsilon_M}{\Delta_{\text{post}}^2}. \quad (\text{C9})$$

The explicit effects

$$B_y = P_{z(y)} M_y P_{z(y)}, \quad D_z = P_z - \sum_{y: z(y)=z} B_y \quad (\text{C10})$$

form an exact block refinement. Trace-norm Cauchy–Schwarz gives a symmetric finite-experiment deficiency bound

$$\delta_{\text{LR}}(M, \widetilde{M}) \leq \min\{1, L(M) + \sqrt{L(M)[d - L(M)]}\}. \quad (\text{C11})$$

For completeness, put $P = P_{z(y)}$, $Q = I - P$, and $L_y = \text{Tr } Q M_y Q$. Then

$$\|M_y - B_y\|_1 \leq L_y + 2\sqrt{(\text{Tr } P M_y P) L_y}. \quad (\text{C12})$$

Summing and using Cauchy–Schwarz once more gives

$$\sum_y \|M_y - B_y\|_1 \leq L(M) + 2\sqrt{L(M)[d - L(M)]}. \quad (\text{C13})$$

To simulate M from $\widetilde{M}$, report y after outcome B_y and distribute each D_z among the labels assigned to block z . In the reverse direction, map outcome y of M to the correspondingly named outcome of $\widetilde{M}$ and never report a D_z label. For every signal state, the two resulting total-variation errors are bounded by one half of the last display plus $L(M)/2$. Taking the larger directed deficiency proves Eq. (C11) (and the trivial upper bound is one). Thus near-optimal measurements are close in their induced statistical experiment to refinements of the canonical posterior PVM, even though arbitrary effect-wise or diamond-norm closeness is neither claimed nor expected.

At a feasible numerical tuple $\widehat{\mathbf{H}}$, let $R_a = \nabla_a \Phi(\widehat{\mathbf{H}})$ and

$$C = \sum_i a_i s_i s_i^\top, \quad r_g^2 = \sum_{a,b} (C^{-1})_{ab} \text{Tr}(R_a R_b). \quad (\text{C14})$$

This is precisely the squared dual norm of the gradient. One-strong concavity implies, for $\mathbf{K} = \mathbf{H}_\star - \widehat{\mathbf{H}}$,

$$\Phi(\mathbf{H}_\star) - \Phi(\widehat{\mathbf{H}}) \leq \langle \nabla \Phi(\widehat{\mathbf{H}}), \mathbf{K} \rangle - \frac{1}{2} \|\mathbf{K}\|_{\text{post}}^2. \quad (\text{C15})$$

Dual Cauchy–Schwarz and maximization of the scalar quadratic give the first bound below; the same strong-concavity inequality together with stationarity at $\mathbf{H}_\star$ gives the second:

$$U_{\text{aff}} \leq \Phi(\widehat{\mathbf{H}}) + \frac{1}{2} r_g^2, \quad \sum_i a_i \|G_i(\widehat{\mathbf{H}}) - G_i^\star\|_F^2 \leq r_g^2. \quad (\text{C16})$$

The first inequality is the binary stopping certificate used in the Letter. A formally computer-assisted certificate additionally requires outward-rounded matrix functions, eigenvalue bounds, and mutual-information evaluation; the reported benchmark uses float64 arithmetic.

Appendix D: Binary construction and singular closure

If one prior vanishes, or if the two states coincide, every measurement has zero mutual information and any rank-one PVM is optimal. We henceforth take $0 < p, q < 1$ and distinct states.

For a faithful binary ensemble $\mathcal{E} = \{(p, \rho_0), (q, \rho_1)\}$, define

$$\begin{aligned} \Phi_{\text{bin}}(H) = & p \text{Tr } \rho_0 \log(I + qH) \\ & + q \text{Tr } \rho_1 \log(I - pH), \end{aligned} \quad (\text{D1})$$

with $I + qH \succ 0$ and $I - pH \succ 0$, and choose

$$\rho_0 = \tau + q(\rho_0 - \rho_1), \quad \rho_1 = \tau - p(\rho_0 - \rho_1). \quad (\text{D2})$$

The affine program is the single-variable maximization of Eq. (D1). Its optimized posteriors are

$$G_0^\star = I + qH_\star, \quad G_1^\star = I - pH_\star, \quad (\text{D3})$$

so they are functions of one generator and commute. Their common spectral PVM is the spectral PVM of H_* , and the posterior-algebra theorem proves that it attains I_{acc} . Refining degenerate spectral projectors to rank one cannot decrease mutual information and therefore preserves optimality.

If one or both states are singular, restrict to $P_0 = \text{supp } \tau$. Because $p, q > 0$ and $\tau = p\rho_0 + q\rho_1$, positivity gives $\text{supp } \rho_i \subseteq P_0$ for both labels. Put

$$\rho_{i,\epsilon} = (1 - \epsilon)\rho_i + \epsilon P_0/d_0, \quad d_0 = \text{rank } P_0. \quad (\text{D4})$$

The perturbed ensemble is faithful on $P_0\mathcal{H}$ and remains binary. Its optimal spectral PVM may be rank-one refined. An ordered rank-one PVM is a d_0 -tuple of mutually orthogonal rank-one projections summing to P_0 ; this set is closed and bounded in finite dimension and hence compact. Thus a sequence $\epsilon_k \downarrow 0$ has a subsequence $P^{(\epsilon_k)} \rightarrow P$. The outcome probabilities converge jointly with the states and PVMs, and continuity of $x \log x$ at zero makes finite-alphabet mutual information jointly continuous. Consequently, for every fixed finite POVM M ,

$$I(\mathcal{E}, P) = \lim_k I(\mathcal{E}_{\epsilon_k}, P^{(\epsilon_k)}) \geq \lim_k I(\mathcal{E}_{\epsilon_k}, M) = I(\mathcal{E}, M). \quad (\text{D5})$$

Davies' theorem then proves attainment for the original singular binary ensemble. Every ambient-space POVM compresses to $P_0\mathcal{H}$ with identical signal statistics. Adjoining any rank-one resolution of $P_0^\perp$ extends the limiting PVM to the original Hilbert space without changing its mutual information.

We finally verify the prior route through Fang–Fawzi–Fawzi. Their finite-dimensional projective-reduction theorem states the following. Let f be convex and lower semicontinuous with $(0, \infty) \subset \text{dom } f$. If an interval J admits a one-to-one parametrization $\psi : J \rightarrow \text{dom } f^*$ for which ψ is operator concave and $f^* \circ \psi$ is operator convex, then the measured f -divergence optimized over POVMs equals the value optimized over PVMs [9, Theorem 2].

If all signals in a collinear ensemble coincide, every measurement has zero mutual information. Otherwise choose two extreme signal states ρ_0 and ρ_1 and write $\rho_i = (1 - \lambda_i)\rho_0 + \lambda_i\rho_1$, where $0 \leq \lambda_i \leq 1$, some $\lambda_i = 0$, some $\lambda_i = 1$, and $\bar{\lambda} = \sum_i p_i \lambda_i \in (0, 1)$. The mutual information of a measurement is the measured f -divergence of the two endpoint outcome distributions for

$$f(t) = \sum_i p_i [(1 - \lambda_i) + \lambda_i t] \log \frac{(1 - \lambda_i) + \lambda_i t}{(1 - \bar{\lambda}) + \bar{\lambda} t}. \quad (\text{D6})$$

Set

$$h = \frac{t - 1}{(1 - \bar{\lambda}) + \bar{\lambda} t}, \quad \psi(h) = \sum_i p_i \lambda_i \log[1 + (\lambda_i - \bar{\lambda})h]. \quad (\text{D7})$$

On $J = (-1/(1 - \bar{\lambda}), 1/\bar{\lambda})$, direct differentiation and Fenchel conjugacy give

$$f'(t) = \psi(h), \quad (\text{D8})$$

$$f^*(\psi(h)) = - \sum_i p_i (1 - \lambda_i) \log[1 + (\lambda_i - \bar{\lambda})h]. \quad (\text{D9})$$

The first expression is operator concave and the second is operator convex. Furthermore,

$$\psi'(h) = (1 - \bar{\lambda}h) \sum_i \frac{p_i (\lambda_i - \bar{\lambda})^2}{1 + (\lambda_i - \bar{\lambda})h} > 0. \quad (\text{D10})$$

The Möbius map sends $t \in (0, \infty)$ bijectively onto J . Since $\psi(h) = f'(t)$ and $\psi' > 0$, one-dimensional Fenchel duality identifies its range with $\text{dom } f^*$. Indeed, the signal with $\lambda_i = 1$ gives $f'(0+) = -\infty$, while the signal with $\lambda_i = 0$ makes the opposite limiting slope inadmissible because $f(t) - tf'(\infty) \rightarrow -\infty$. Thus no endpoint is omitted. With the lower-semicontinuous conventions at ratios zero and infinity, all hypotheses of Fang–Fawzi–Fawzi's theorem hold. Therefore the POVM and PVM suprema agree; rank-one refinement and compactness give attainment. This route establishes prior logical coverage of projective sufficiency. It does not identify the canonical posterior algebra, classify every optimizer, or give the rigidity and finite-iterate certificates developed here.

Appendix E: Posterior-spectral receiver and numerical benchmarks

For a faithful binary ensemble $\mathcal{E} = \{(p, \rho_0), (q, \rho_1)\}$, $q = 1 - p$, the affine coordinates are $s_0 = q$ and $s_1 = -p$, and the posterior program is Eq. (D1). Its unique optimizer generates both posterior operators. The spectral PVM of $H_\star$ is therefore information-optimal. At a feasible iterate $\hat{H}$, intrinsic strong concavity gives

$$I_{\text{acc}} \leq U_{\text{aff}} \leq \Phi(\hat{H}) + \frac{1}{2}r_g^2, \quad (\text{E1})$$

where r_g is the dual norm of the gradient in the intrinsic posterior geometry. The difference between this upper bound and the directly evaluated spectral-PVM information is the stopping certificate used below.

Implementation of the posterior-spectral receiver

For the binary coordinates $s_0 = q$ and $s_1 = -p$, the intrinsic curvature matrix is the scalar

$$C = p^2 q^2 [p\lambda_{\min}(\rho_0) + q\lambda_{\min}(\rho_1)] > 0, \quad (\text{E2})$$

and the gradient of Eq. (D1) is

$$R(H) = pq [\mathcal{L}_{I+qH}(\rho_0) - \mathcal{L}_{I-pH}(\rho_1)]. \quad (\text{E3})$$

Intrinsic strong concavity gives $U_{\text{aff}} - \Phi(H) \leq \text{Tr } R(H)^2 / (2C)$. A backtracking line search preserves $I + qH \succ 0$ and $I - pH \succ 0$; the implementation uses inverse BFGS, although the certificate is independent of the proposal method. Matrix logarithms and Fréchet derivatives are evaluated in an eigenbasis using stable divided differences. A formally computer-assisted certificate replaces floating-point values by outward-rounded enclosures. For singular states, one first compresses to suppress τ and introduces a validated depolarization error.

Algorithms and benchmark protocol

We use a fixed-seed suite containing 408 stress instances: 24 cases at every dimension or conditioning point. It spans dimensions $d = 2, 4, 8, 16$, signal-state spectral floors down to 10^{-6} , priors up to 0.99, and nearly coincident state pairs $\rho_1 = (1 - \delta)\rho_0 + \delta\sigma$ with δ down to 10^{-2} .

The compared methods are as follows.

1. *Posterior-spectral receiver*: feasibility-preserving inverse BFGS applied to Eq. (D1), followed by spectral decomposition and Eq. (E1).
2. *Direct POVM ascent*: a multistart, SOMIM-style mutual-information ascent over d^2 POVM effects, using multiplicative positive updates and POVM renormalization [18]. Starts are uniform, Helstrom, and PGM.
3. *Multistart PVM*: BFGS on $U(d)$ in local exponential coordinates, initialized by identity, Helstrom, and Haar-random bases.
4. *Helstrom and PGM*: the minimum-error spectral measurement and the square-root measurement, respectively [1, 19].

The strong baselines optimize the same mutual information but have no independent global upper bound. In the stress suite the optimization budgets are deliberately fixed: at most 300 posterior iterations, 100 accepted direct-POVM iterations per deterministic start, and 30 PVM-BFGS iterations per start. Stress-test regret therefore measures finite-resource robustness, not a difference in the mathematical optimum.

Solver profiles

The main-text solver-profile figure uses all 408 stress instances. Panel (a) is an accuracy profile: for every target regret ϵ it reports the fraction of instances meeting the target. Panel (b) is a time-to-accuracy profile at $\epsilon = 10^{-6}$ nat. The posterior-spectral receiver reaches the target on all 408 cases, with median successful time 12.9 ms. Multistart

PVM and direct POVM ascent reach the target on 45.8% and 4.7% of cases; Helstrom and PGM reach it on 0.25% and 0%. The black curve uses the affine upper-minus-PVM gap, whereas the remaining curves use empirical regret relative to the posterior-spectral PVM.

All numerical certificate values are float64 evaluations of exact-arithmetic bounds, not outward-rounded interval enclosures. Timings are implementation- and machine-dependent. Complete CSV files, fixed seeds, and plotting scripts are included in the `anc/` directory of the arXiv source archive.

Appendix F: Scope, prior work, and limitations

The existence of a projective optimum for finite collinear ensembles is also a specialization of the measured- f projective-reduction theorem of Fang, Fawzi, and Fawzi [9]. The posterior-algebra construction is a proof and adds the canonical measurement, complete optimizer classification, robust block rigidity, and certified finite-iterate design. Classical minimal sufficiency and postprocessing equivalence motivate the Blackwell interpretation [16, 17].

The faithful-state rigidity constants deteriorate near singular boundaries. Existence of the binary optimum extends to arbitrary singular states by support compression, depolarization, and compactness, but a uniform ambient Frobenius-rigidity theorem does not. States faithful on a common support are covered by restricting every operator to that support. Varying-support pure states require a visibility-weighted metric. The benchmark does not prove polynomial bit complexity or asymptotic scaling exponents; it demonstrates the finite-resource consequence of the one-generator reduction.